ic_autolab round 33 (generic track): report

Protocol: PROTOCOL.md, frozen 2026-11-04T12:09:43Z at git `aea9539aec5b52da765ca2ebd1c2afe894a64d45`; route `local`. Counted GAE only (deterministic, host independent); no wall-time claim; development loop, no validation-gate row. Loop home: `../`; ledger: `../ledger.json`.

Claim record: `PCC-20261104-c28f16` (`python3 tools/pcc/pcc.py standing PCC-20261104-c28f16`). Until a signed replay and recompute exist, this round's outcome is an unverified assertion [pcc:unverified].

Outcome

k	class	sel	sel / inc at w = 0 [95%]	w = 0.1	w = 1	band	S sel (w = 0 / 0.1 / 1)	S inc	S rho-neg	distance sel / rho
20	**no_challenger**	none (best candidate `wwalkGg-x10-n2`, train S 2.595 / 2.654 / 3.187)	-	-	-	-	-	2.541 / 2.654 / 3.673	1.576 / 1.690 / 2.718	inc: 1.6x / 1.6x / 1.4x
24	**improvement**	wwalkGg-x09-n2	0.721 [0.610, 0.853]	0.711 [0.595, 0.850]	0.647 [0.502, 0.834]	sel < inc	1.673 / 1.718 / 2.117	2.321 / 2.416 / 3.270	1.171 / 1.276 / 2.225	1.4x / 1.3x / 1.0x
28	**improvement**	wwalkGg-x09-n2	0.795 [0.673, 0.940]	0.770 [0.647, 0.918]	0.630 [0.501, 0.792]	sel < inc	1.572 / 1.611 / 1.961	1.977 / 2.091 / 3.113	1.052 / 1.151 / 2.039	1.5x / 1.4x / 1.0x
32	**improvement**	wwalkG-x06-n2	0.715 [0.648, 0.788]	0.697 [0.626, 0.777]	0.620 [0.528, 0.728]	sel < inc	1.515 / 1.625 / 2.608	2.120 / 2.329 / 4.207	0.975 / 1.071 / 1.933	1.6x / 1.5x / 1.3x

Classes follow the protocol's decision rule. Distance to rho is the distance remaining, not a gate. The reasons per size:

- k = 20: no_challenger
- k = 24: compare ops.status = ok, admissible = true; sel / inc = 0.7210 [0.610, 0.853] at w = 0; band: sel < inc; A/A inc-aa / inc interval [1.000, 1.000] contains 1
- k = 28: compare ops.status = ok, admissible = true; sel / inc = 0.7950 [0.673, 0.940] at w = 0; band: sel < inc; A/A inc-aa / inc interval [1.000, 1.000] contains 1
- k = 32: compare ops.status = ok, admissible = true; sel / inc = 0.7146 [0.648, 0.788] at w = 0; band: sel < inc; A/A inc-aa / inc interval [1.000, 1.000] contains 1

Train (development screen, in sample)

k = 20: session `ECBS1h66ff64289214` (run exit 0, verify exit 0, receipt sha256 `ef7b2fd4780fe2bbe16d7f08240698196e2f3bac0892b7ce8c561e105a9cb7cd`)

arm	family	N factor	base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
`rho-neg`	rho	-	-	`rho.negation`	32	32	1.576 / 1.690 / 2.718	matched reference
`inc`	wsum2g-b15	2	`prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=15,free=1`	32	32	2.541 / 2.654 / 3.673	incumbent
`wwalkG-x06-n2`	wwalkG-x06	2	`prime-projected-abscissa:size=2,scan_cap=32`	`wwalk:x=0.6,g=1,grow=2`	32	32	5.491 / 5.604 / 6.619	exploration: family wwalkG-x06 untried at k = 20, N = 0.0001 x N_a
`wwalkGg-x06-n2`	wwalkGg-x06	2	`prime-projected-abscissa:size=2,scan_cap=32`	`wwalk:x=0.6,g=1,grow=2`	32	32	3.311 / 3.370 / 3.901	exploration: family wwalkGg-x06 untried at k = 20, N = 0.0001 x N_a
`wwalkG-x07-n2`	wwalkG-x07	2	`prime-projected-abscissa:size=2,scan_cap=32`	`wwalk:x=0.7,g=1,grow=2`	32	32	5.266 / 5.338 / 5.993	exploration: family wwalkG-x07 untried at k = 20, N = 0.0001 x N_a
`wwalkGg-x07-n2`	wwalkGg-x07	2	`prime-projected-abscissa:size=2,scan_cap=32`	`wwalk:x=0.7,g=1,grow=2`	32	32	3.105 / 3.145 / 3.503	exploration: family wwalkGg-x07 untried at k = 20, N = 0.0001 x N_a
`wwalkG-x08-n2`	wwalkG-x08	2	`prime-projected-abscissa:size=2,scan_cap=32`	`wwalk:x=0.8,g=1,grow=2`	32	32	5.542 / 5.600 / 6.124	exploration: family wwalkG-x08 untried at k = 20, N = 0.0001 x N_a

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
wwalkGg-x08-n2	wwalkGg-x08	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=0.8,g=1,grow=2	32	32	3.326 / 3.352 / 3.591	exploration: family wwalkGg-x08 untried at k = 20, N = 0.0001 x N_a
wwalkG-x09-n2	wwalkG-x09	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=0.9,g=1,grow=2	32	32	5.728 / 5.793 / 6.371	exploration: family wwalkG-x09 untried at k = 20, N = 0.0001 x N_a
wwalkGg-x09-n2	wwalkGg-x09	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=0.9,g=1,grow=2	32	32	3.478 / 3.506 / 3.760	exploration: family wwalkGg-x09 untried at k = 20, N = 0.0001 x N_a
wwalkG-x10-n2	wwalkG-x10	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=1.0,g=1,grow=2	32	32	4.654 / 4.743 / 5.541	exploration: family wwalkG-x10 untried at k = 20, N = 0.0001 x N_a
wwalkGg-x10-n2	wwalkGg-x10	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=1.0,g=1,grow=2	32	32	2.595 / 2.654 / 3.187	exploration: family wwalkGg-x10 untried at k = 20, N = 0.0001 x N_a
wwalkG-x12-n2	wwalkG-x12	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=1.2,g=1,grow=2	32	32	4.885 / 4.977 / 5.799	exploration: family wwalkG-x12 untried at k = 20, N = 0.0001 x N_a
wwalkGg-x12-n2	wwalkGg-x12	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=1.2,g=1,grow=2	32	32	2.626 / 2.667 / 3.040	exploration: family wwalkGg-x12 untried at k = 20, N = 0.0001 x N_a

k = 24: session `ECBS1hd3cd3286deb9` (run exit 0, verify exit 0, receipt sha256 `1f46a35e572ff83f8e3c143a7ec3e2390e52445473ad875c7d23e21e00344d7c`)

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
rho-neg	rho	- -	rho.negation	32	32	1.152 / 1.255 / 2.185	matched reference
inc	wwalkg-x08-grow	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=0.8,grow=1	32	32	2.327 / 2.422 / 3.281	incumbent
wwalkG-x06-n2	wwalkG-x06	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=0.6,g=1,grow=2	32	32	3.146 / 3.232 / 4.013	exploration: family wwalkG-x06 untried at k = 24, N = 0.0001 x N_a
wwalkGg-x06-n2	wwalkGg-x06	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=0.6,g=1,grow=2	32	32	2.282 / 2.308 / 2.545	exploration: family wwalkGg-x06 untried at k = 24, N = 0.0001 x N_a
wwalkG-x07-n2	wwalkG-x07	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=0.7,g=1,grow=2	32	32	2.441 / 2.517 / 3.198	exploration: family wwalkG-x07 untried at k = 24, N = 0.0001 x N_a
wwalkGg-x07-n2	wwalkGg-x07	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=0.7,g=1,grow=2	32	32	1.847 / 1.894 / 2.322	exploration: family wwalkGg-x07 untried at k =

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
							24, N = 0.0001 x N_a
wwalkG-x08-n2	wwalkG-x08	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=0.8,g=1,grow=2	32	32	3.188 / 3.227 / 3.581	exploration: family wwalkG-x08 untried at k = 24, N = 0.0001 x N_a
wwalkGg-x08-n2	wwalkGg-x08	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=0.8,g=1,grow=2	32	32	2.586 / 2.614 / 2.867	exploration: family wwalkGg-x08 untried at k = 24, N = 0.0001 x N_a
wwalkG-x09-n2	wwalkG-x09	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=0.9,g=1,grow=2	32	32	2.554 / 2.678 / 3.799	exploration: family wwalkG-x09 untried at k = 24, N = 0.0001 x N_a
wwalkGg-x09-n2	wwalkGg-x09	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=0.9,g=1,grow=2	32	32	1.685 / 1.730 / 2.140	exploration: family wwalkGg-x09 untried at k = 24, N = 0.0001 x N_a
wwalkG-x10-n2	wwalkG-x10	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=1.0,g=1,grow=2	32	32	3.595 / 3.643 / 4.077	exploration: family wwalkG-x10 untried at k = 24, N = 0.0001 x N_a
wwalkGg-x10-n2	wwalkGg-x10	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=1.0,g=1,grow=2	32	32	2.856 / 2.875 / 3.045	exploration: family wwalkGg-x10 untried at k = 24, N = 0.0001 x N_a
wwalkG-x12-n2	wwalkG-x12	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=1.2,g=1,grow=2	32	32	2.615 / 2.689 / 3.358	exploration: family wwalkG-x12 untried at k = 24, N = 0.0001 x N_a
wwalkGg-x12-n2	wwalkGg-x12	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=1.2,g=1,grow=2	32	32	1.967 / 2.009 / 2.385	exploration: family wwalkGg-x12 untried at k = 24, N = 0.0001 x N_a

k = 28: session `ECBS1h055f1cf0371f` (run exit 0, verify exit 0, receipt sha256 `3d840a2a354f30b1892b9b5ae1efba4a4bacd878698a3db00483b88ae00197db`)

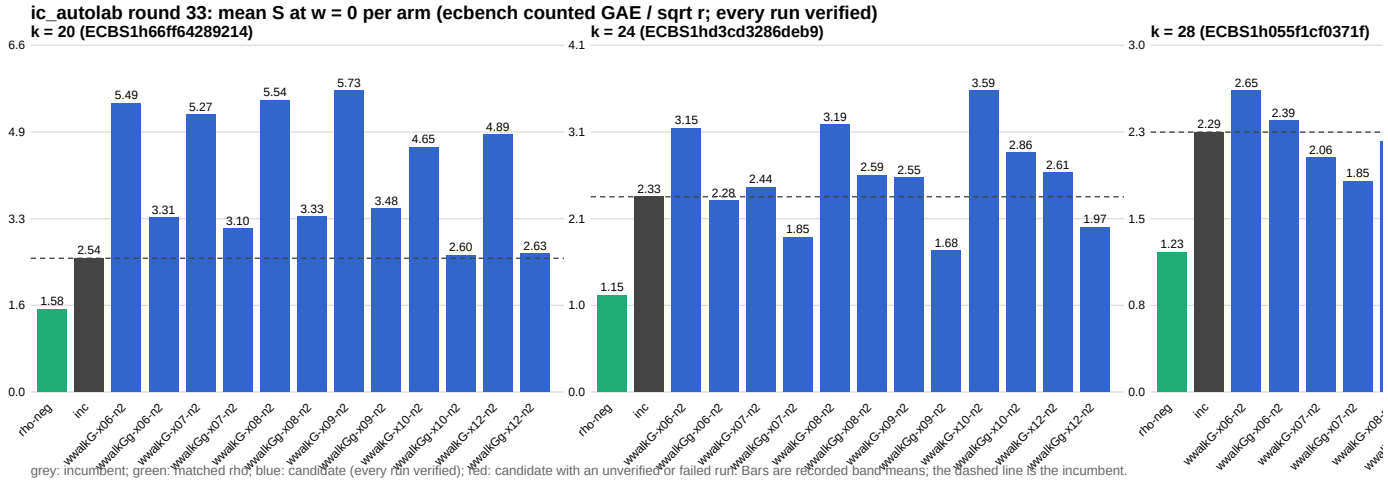
arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
rho-neg	rho	- -	rho.negation	32	32	1.231 / 1.347 / 2.397	matched reference
inc	wwalkg-x08	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=0.8	32	32	2.287 / 2.431 / 3.732	incumbent
wwalkG-x06-n2	wwalkG-x06	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=0.6,g=1,grow=2	32	32	2.652 / 2.762 / 3.746	exploration: family wwalkG-x06 untried at k = 28, N = 0.0001 x N_a
wwalkGg-x06-n2	wwalkGg-x06	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=0.6,g=1,grow=2	32	32	2.385 / 2.434 / 2.878	exploration: family wwalkGg-x06 untried at k = 28, N = 0.0001 x N_a
wwalkG-x07-n2	wwalkG-x07	2 prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=0.7,g=1,grow=2	32	32	2.063 / 2.135 / 2.789	exploration: family wwalkG-x07

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
							untried at k = 28, N = 0.0001 x N_a
wwalkGg-x07-n2	wwalkGg-x07	2	prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=0.7,g=1,grow=2	32	32	1.852 / 1.888 / 2.215	exploration: family wwalkGg-x07 untried at k = 28, N = 0.0001 x N_a
wwalkG-x08-n2	wwalkG-x08	2	prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=0.8,g=1,grow=2	32	32	2.203 / 2.272 / 2.897	exploration: family wwalkG-x08 untried at k = 28, N = 0.0001 x N_a
wwalkGg-x08-n2	wwalkGg-x08	2	prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=0.8,g=1,grow=2	32	32	1.935 / 1.964 / 2.224	exploration: family wwalkGg-x08 untried at k = 28, N = 0.0001 x N_a
wwalkG-x09-n2	wwalkG-x09	2	prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=0.9,g=1,grow=2	32	32	1.990 / 2.109 / 3.178	exploration: family wwalkG-x09 untried at k = 28, N = 0.0001 x N_a
wwalkGg-x09-n2	wwalkGg-x09	2	prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=0.9,g=1,grow=2	32	32	1.644 / 1.690 / 2.106	exploration: family wwalkGg-x09 untried at k = 28, N = 0.0001 x N_a
wwalkG-x10-n2	wwalkG-x10	2	prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=1.0,g=1,grow=2	32	32	1.983 / 2.045 / 2.609	exploration: family wwalkG-x10 untried at k = 28, N = 0.0001 x N_a
wwalkGg-x10-n2	wwalkGg-x10	2	prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=1.0,g=1,grow=2	32	32	1.813 / 1.848 / 2.166	exploration: family wwalkGg-x10 untried at k = 28, N = 0.0001 x N_a
wwalkG-x12-n2	wwalkG-x12	2	prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=1.2,g=1,grow=2	32	32	2.130 / 2.250 / 3.326	exploration: family wwalkG-x12 untried at k = 28, N = 0.0001 x N_a
wwalkGg-x12-n2	wwalkGg-x12	2	prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=1.2,g=1,grow=2	32	32	1.807 / 1.856 / 2.295	exploration: family wwalkGg-x12 untried at k = 28, N = 0.0001 x N_a

k = 32: session `ECBS1h866cd7639cb7` (run exit 0, verify exit 0, receipt sha256 `f3ecad56a9d64c014b65247ec2fb833cb7611356d0bd1d88512c4081d17ca1e3`)

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
rho-neg	rho	- -	rho.negation	32	32	0.839 / 0.922 / 1.662	matched reference
inc	wwalk-x10	2	prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=1.0	32	32	1.983 / 2.164 / 3.794	incumbent
wwalkG-x06-n2	wwalkG-x06	2	prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=0.6,g=1,grow=2	32	32	1.336 / 1.409 / 2.071	exploration: family wwalkG-x06 untried at k = 32, N = 0.0001 x N_a
wwalkGg-x06-n2	wwalkGg-x06	2	prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=0.6,g=1,grow=2	32	32	1.375 / 1.421 / 1.842	exploration: family

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
							wwalkGg-x06 untried at k = 32, N = 0.0001 x N_a
wwalkG-x07-n2	wwalkG-x07	2	prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=0.7,g=1,grow=2	32	32	1.701 / 1.773 / 2.425	exploration: family wwalkG-x07 untried at k = 32, N = 0.0001 x N_a
wwalkGg-x07-n2	wwalkGg-x07	2	prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=0.7,g=1,grow=2	32	32	1.608 / 1.641 / 1.937	exploration: family wwalkGg-x07 untried at k = 32, N = 0.0001 x N_a
wwalkG-x08-n2	wwalkG-x08	2	prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=0.8,g=1,grow=2	32	32	1.557 / 1.637 / 2.357	exploration: family wwalkG-x08 untried at k = 32, N = 0.0001 x N_a
wwalkGg-x08-n2	wwalkGg-x08	2	prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=0.8,g=1,grow=2	32	32	1.566 / 1.613 / 2.036	exploration: family wwalkGg-x08 untried at k = 32, N = 0.0001 x N_a
wwalkG-x09-n2	wwalkG-x09	2	prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=0.9,g=1,grow=2	32	32	2.255 / 2.302 / 2.717	exploration: family wwalkG-x09 untried at k = 32, N = 0.0001 x N_a
wwalkGg-x09-n2	wwalkGg-x09	2	prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=0.9,g=1,grow=2	32	32	2.290 / 2.323 / 2.616	exploration: family wwalkGg-x09 untried at k = 32, N = 0.0001 x N_a
wwalkG-x10-n2	wwalkG-x10	2	prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=1.0,g=1,grow=2	32	32	1.683 / 1.759 / 2.440	exploration: family wwalkG-x10 untried at k = 32, N = 0.0001 x N_a
wwalkGg-x10-n2	wwalkGg-x10	2	prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=1.0,g=1,grow=2	32	32	1.718 / 1.766 / 2.193	exploration: family wwalkGg-x10 untried at k = 32, N = 0.0001 x N_a
wwalkG-x12-n2	wwalkG-x12	2	prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=1.2,g=1,grow=2	32	32	2.077 / 2.151 / 2.811	exploration: family wwalkG-x12 untried at k = 32, N = 0.0001 x N_a
wwalkGg-x12-n2	wwalkGg-x12	2	prime-projected-abscissa:size=2,scan_cap=32	wwalk:x=1.2,g=1,grow=2	32	32	1.876 / 1.899 / 2.103	exploration: family wwalkGg-x12 untried at k = 32, N = 0.0001 x N_a



grey: incumbent; green: matched rho; blue: candidate (every run verified); red: candidate with an unverified or failed run. Bars are recorded band means; the dashed line is the incumbent.

train screen

Selection

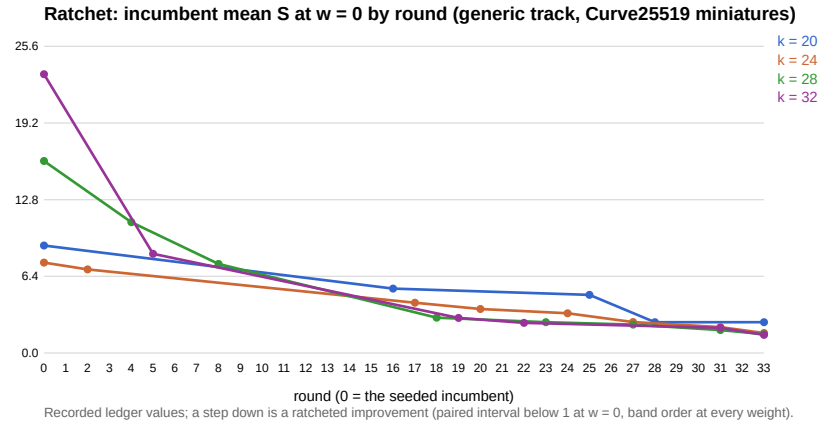
- k = 20: no_challenger (Pareto alternative at w = 1: `wwalkGg-x12-n2`)
- k = 24: challenger `wwalkGg-x09-n2` (Pareto alternative at w = 1: `wwalkGg-x09-n2`)
- k = 28: challenger `wwalkGg-x09-n2` (Pareto alternative at w = 1: `wwalkGg-x09-n2`)
- k = 32: challenger `wwalkG-x06-n2` (Pareto alternative at w = 1: `wwalkGg-x06-n2`)

Holdout sessions

k	session	run exit	verify exit	receipt sha256	levels
24	`ECBS1h05c72414c6d4`	0	0	`43de56f9cde7f622ed22f2e0b24b44b140d2ff22a101f9a64fef6acd568b93a7`	{“verified”:480}
28	`ECBS1hba478ffc1d23`	0	0	`3435de2920cad08d48385589d5ac7048b6a179315ed97a2abf2eeb1de3685281`	{“verified”:480}
32	`ECBS1h8b22612ef731`	0	0	`71c7bf0c4bcbb0e46f155285b7c71dc81d7ce125bf0e79a18fa49f90fa95407c`	{“verified”:480}

Ledger after this round

k	incumbent	since round	S (w = 0 / 0.1 / 1)	distance to rho
20	`wsum2g-b15-n2` (`prime-projected-abscissa:size=2,scan_cap=32` + `wsum:coef=15,free=1`)	28	2.575 / 2.691 / 3.741	1.8x / 1.7x / 1.5x
24	`wwalkGg-x09-n2` (`prime-projected-abscissa:size=2,scan_cap=32` + `wwalk:x=0.9,g=1,grow=2`)	33	1.673 / 1.718 / 2.117	1.4x / 1.3x / 1.0x
28	`wwalkGg-x09-n2` (`prime-projected-abscissa:size=2,scan_cap=32` + `wwalk:x=0.9,g=1,grow=2`)	33	1.572 / 1.611 / 1.961	1.5x / 1.4x / 1.0x
32	`wwalkG-x06-n2` (`prime-projected-abscissa:size=2,scan_cap=32` + `wwalk:x=0.6,g=1,grow=2`)	33	1.515 / 1.625 / 2.608	1.6x / 1.5x / 1.3x



Recorded ledger values; a step down is a ratcheted improvement (paired interval below 1 at w = 0, band order at every weight).

ratchet

Evidence and deviations

- ecbench binary sha256: `737fb5b0901bda2c34a6a059032aa91fb76b1cfe656a4146c21c9778c5127c2e`; `--cpus auto`, allow_busy = true; `verify --replay 8` per session; audit receipts in `analysis/`; sessions in `sessions/` (records in the artifact store when large).
- Route: the local route is an explicitly requested development screen, not the Iso Lab EKS default; counts are deterministic and replayed, wall time is not claimed. No row of this round enters the IC leaderboard or `benchmarks/README.md` (validation gate).
- Train selection is in sample (disclosed); the holdout is fresh (new `target_seed`, excluded exposures). Failed, timed-out and regressing arms are retained above.